

Hoang Minh Thanh

3D Avatar Developer

[hmthanh.github.io](https://github.com/hmthanh) [hmthanh](https://github.com/hmthanh) [in hmthanh](https://www.linkedin.com/in/hmthanh) [Vietnam](#) hmthanhgm@gmail.com

With a strong foundation in real-time animation, Unreal Engine, and generative AI, proficient in WebRTC, Three.js, Unity, and GPU-accelerated rendering pipelines. And experienced in AI-based character perception, motion synthesis, and expressive avatar behavior design. I believe I am well-suited for the 3D Avatar Developer position at **Nudgyt**.

★ Keypoint

- **Proficient in Three.js, WebGL and Web Technologies:** Hands-on experience in optimization scalable production 3D web assets. Vast experience of multiple React web project: ([MillionScope](#), [MetaPet](#)), [openhuman-ai/renderengine](#), [Skeleton Animation](#), ... Proposed the [DeepGesture](#) model. Creator of [OpenHuman](#): [openhuman-ai/renderengine](#), [GLSL Shader](#), [DeepGesture video render](#)
- **Skilled in Blender & Maya:** Experienced in modeling, rigging, texturing, optimizing, and profound understanding [GLTF JSON](#), Structure for exporting web assets. [Unity render](#), [Blender render](#).
Extensive experience in modeling, rigging, texturing, optimization, and 3D assets technical workflows, ZWrap (transfer topology), ZBrush (layer baking). Game made by Unity - (Play my game [Pucca Runner](#)). Hands-on experience in optimization scalable production 3D web assets.
- **Comprehensive understanding Computer Graphics:** *Keyword learned:* Ray Tracing in One Weekend, Subsurface scattering, Rendering Equation, [pbrt-v4](#), [webgl2fundamentals](#), Physics/Spectrum/Eye Vision Structure of Color, ... [My Fillament doc remake](#) (BRDF, Diffuse, Reflection).

🎓 Education

AS	University of Science - VNUHCM, Information Technology	2012 – 2015
BS	University of Science - VNUHCM, Computer Science	Sept. 2018 – Sept. 2020
	• GPA: 3.3/4.0 • Thesis: GCAT - Link Prediction in Knowledge Graph, [Report], [Code]	
MSc	University of Science - VNUHCM, Computer Science	Oct. 2021 – Dec. 2024
	• GPA: 3.44/4.0 • Thesis: OpenHuman: A conversational gesture synthesis system based on emotions and semantics, [Report], [Code]	

📖 Publications

DeepGesture: A conversational gesture synthesis system based on emotions and semantics [arXiv] , [Homepage]	Jul. 2025
Towards a GENE Leaderboard [arXiv]	Oct. 2024
Graph Collaborative Attention Network for Link Prediction in Knowledge Graphs [arXiv]	Oct. 2020

💼 Experience

Rakumo Inc, Software Engineer	Jul. 2019 – Apr. 2020
• Developed a Django-based web application with integrated DocuSign.	
FPT Software, AWS Data Engineer	Oct. 2020 – Nov. 2021
• CI/CD pipeline terraform in Jenkins, implemented and modified real-time and batch	

data pipelines on AWS using Kinesis, Lambda, S3, Glue, StepFunction, SNS, ..etc..

- Cox Automotive (onsite-offshore): Collaborated with international teams in Agile

VNG Corp, Software Engineer

Oct. 2021 – May 2024

ZDN Team (Zalo Content Delivery Network): caching service serve millions request

- Implement scheduled `ScheduledThreadPool` for monitoring stats execution.
- Built Ant admin dashboard with server visualization via amchart, p5.js in NextJS.
- Upgrade internal Golang package for Apache Thrift

OpenHuman [🔗](https://openhuman.ai) (openhuman.ai [🔗](https://openhuman.ai)), Creator

May 2024 – Present

- Join GENE Research Team ([GENEA Leaderboard](#) [🔗](#), [HEMVIP](#) [🔗](#)): Designed and built a full-stack, production-ready leaderboard system to evaluate multiple AI gesture generation models, integrating user study workflows via Prolific.
- [DeepFACS](#) [🔗](#): Create multiple blendshape compatible with ARKit 52 for blendshape, using ZWrap for transfer topology from 3DScanStore basemesh to facial expression. Digitized myself using Character Creator 4, [Universal Human](#) [🔗](#).
- [DeepGesture](#) [🔗](#) ([Demo](#) [🔗](#)): Developed gesture generation AI system on Unity based on the [DeepPhase](#) [🔗](#). Retarget skeleton animation using MotionBuilder.

</> Selected Projects

DeepGesture ([Code](#) [🔗](#))

deepgesture.github.io [🔗](#)

Three.js realistic face for OpenHuman ([Code](#) [🔗](#))

renderengine.pages.dev [🔗](#)

- Loaded and scaled facial OBJ parts in Blender (eyeball, head, tongue, etc.); applied edited albedo, roughness, specular, clearcoat maps; configured dat.GUI, environment toggle, and tone mapping for realistic facial rendering

HairMimic - In progress hair simulation for OpenHuman ([Code](#) [🔗](#))

hairmimic.github.io [🔗](#)

- Generate hair from Maya XGen Grooming and load hair meshes, hair strand design.

MillionScope ([Code](#) [🔗](#)) ChatGPT-Style with Chat SDK. Stream response result of Cloudflare Workers & Cloudflare AI.

millionscope.com [🔗](#)

Unity Game Pucca Runner ([Code](#) [🔗](#)) Pathfinding with a 2D hash map and Dijkstra's algorithm; runner moves toward the player and triggers collision on boundary breach

[Pucca Runner](#) [🔗](#)

Three.js Demo Project: [GlowShader](#) [🔗](#) / [Code](#) [🔗](#), [SpringSphere](#) [🔗](#) / [Code](#) [🔗](#), [Vinfast Car](#) [🔗](#) / [Code](#) [🔗](#), [Jumming Girl](#) [🔗](#) / [Code](#) [🔗](#), [Cross Sphere](#) [🔗](#) / [Code](#) [🔗](#).

Others projects: [Morph target animation ReadyPlayerMe](#) [🔗](#), [MetaPet](#) (metapet.vercel.app [🔗](#)), cortexpod.com [🔗](#); pi-thagon.com [🔗](#), More on: github.com/hmthanh [🔗](#)

🌟 Certifications

- [Critical Thinking Certification 2014](#) [🔗](#), [Emotion Recognition Certification 2019](#) [🔗](#), [Summer ML](#) [🔗](#)
- Coursera: [Design Pattern](#) [🔗](#), [TensorFlow Developer Professional](#) [🔗](#) 2021
- **AWS: Solutions Architect - Associate** [🔗](#), **Machine Learning - Specialty** [🔗](#); **Solution Architect** [🔗](#) 2021

Technologies

3D Engines & Tools: Unreal Engine, Metahuman, Unity, Blender, Maya, ZBrush, MotionBuilder

Avatar Systems: ARKit 52 Blendshape, SMPL, ZWrap, Facial Rigging

Rendering: Three.js, WebGL, GLSL Shader, Real-time Material Mapping

Realtime Systems: WebRTC, Cloudflare Workers, Apache Thrift